

Simultaneous Multi-Way Alignment of LoRA Ensembles via Generalized Procrustes Analysis

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Abstract

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Acknowledgements

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Chapter 1

Introduction

The past years have witnessed a fundamental shift in how task-specific natural language processing systems are built. The emergence of large pre-trained language models (Brown et al., 2020) has established a dominant paradigm in which a single, general-purpose model is adapted to downstream tasks through fine-tuning. However, full fine-tuning of models with tens or hundreds of billions of parameters is prohibitively expensive in both compute and memory. Parameter-efficient fine-tuning (PEFT) methods (Houlsby et al., 2019; Li and Liang, 2021; Lester, Al-Rfou and Constant, 2021) have addressed this constraint by confining updates to a small subset of parameters. Among these, Low-Rank Adaptation (LoRA) (Hu et al., 2022) has emerged as the dominant approach, and its quantised variant QLoRA (Dettmers et al., 2023) has extended access to fine-tuning to settings with severely limited hardware. By injecting pairs of low-rank matrices into frozen Transformer layers and training only these, LoRA enables task-specific adaptation at a fraction of the cost of full fine-tuning.

This proliferation of cheap, specialised adapters has created a new problem: an organisation or practitioner may accumulate an ensemble of LoRA modules, each demonstrably strong on its own task, but requiring separate model instances at inference time. Deploying N task-specific adapters individually incurs N -fold memory and serving overhead. The natural question is whether these specialised adapters can be *combined* into a single multi-task adapter without any further training.

Model merging offers a compelling answer: combine independently fine-tuned models by operating directly in weight space, requiring neither retraining nor access to the original training data (Wortsman et al., 2022; Ilharco et al., 2022). Work on Model Soups (Wortsman et al., 2022) established that averaging the weights of models fine-tuned from a shared initialisation can improve accuracy and robustness. Task Arithmetic (Ilharco et al., 2022) formalised this idea by defining *task vectors*, the signed difference between fine-tuned and pre-trained weights, and demonstrating that these vectors can be composed arithmetically. These results rest on a remarkable geometric property of the fine-tuning loss landscape: models trained from the same pre-trained checkpoint tend to remain within a single low-error basin (Neyshabur, Sedghi and Zhang, 2020), making linear interpolation in weight space well-posed. Yet naive weight averaging fails systematically when independently trained parameters encode similar representations in *different* coordinate systems. When this occurs, averaging causes destructive interference: corresponding features cancel rather than reinforce, collapsing multi-task performance. TIES-Merging (Yadav et al., 2023) and DARE (Yu et al., 2024) have proposed mitigations based on sparsification and sign resolution, but both treat

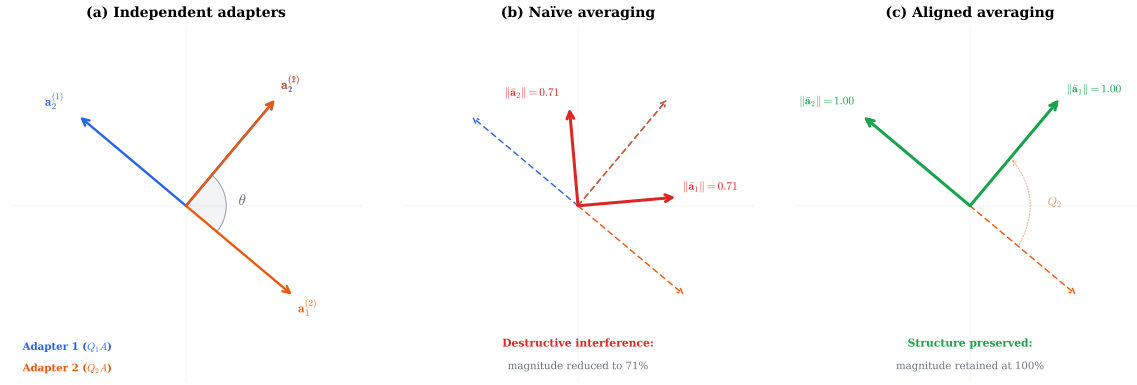


Figure 1.1: Illustration of the rotational misalignment problem in a simplified two-dimensional setting. (a) Two adapters (blue, orange) learn rotated versions of the same subspace; the angle θ between their principal directions is the misalignment to be resolved. (b) Naïve averaging of misaligned unit vectors produces a vector of magnitude $\sqrt{2}/2 \approx 0.71$, a 29% loss due to destructive interference. (c) GPA finds per-adapter orthogonal rotations Q_1, Q_2 that bring both bases into a shared coordinate system; the aligned vectors are collinear, so averaging is fully constructive and magnitude is preserved at 1.00. In practice, each $Q_i \in \mathbb{R}^{r \times r}$ is an orthogonal matrix found in closed form via SVD, where r is the LoRA rank.

interference as a *statistical* problem of conflicting parameter values. The geometric root cause remains unaddressed: the low-rank subspaces that independently trained adapters span may be meaningfully *rotated* relative to one another, and no amount of sign resolution or sparsification can undo a rotation.

The factored structure of LoRA makes this geometric challenge particularly transparent. Each adapter parameterises its weight update as $\Delta W = BA$, where $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ with rank $r \ll \min(d_{\text{in}}, d_{\text{out}})$. The rows of A span the input subspace from which task-relevant features are projected; B maps those projections back to the output space. Independent fine-tuning offers no guarantee that the A matrices of different adapters will span the same coordinate system: each optimisation trajectory is free to choose any rotation of a shared latent subspace. This is not a pathology but an inevitable consequence of the rotational symmetry of the low-rank parameterisation. Figure 1.1 illustrates this problem in a simplified two-dimensional setting: when two adapters learn rotated versions of the same subspace, naïve averaging of the basis vectors causes destructive cancellation, whereas aligning the bases before averaging preserves the shared structure. Preliminary analysis of adapters trained on the GLUE benchmark using Qwen2.5-1.5B with QLoRA confirms this picture quantitatively: the Frobenius norms of B factors differ by over $8\times$ between adapters trained on MNLI and RTE (Figure 1.2), and naïve factor-space merging at $\lambda = 1.0$ causes universal collapse on the CoLA task regardless of the merging strategy, with every method producing a Matthews correlation of exactly 0.0 despite the single-task adapter achieving 0.6522. These observations motivate a geometric rather than statistical approach to the interference problem.

The key insight is that alignment should precede merging. If orthogonal rotations $\{Q_i\}$ can be found such that $\{Q_i A_i\}$ all span approximately the same coordinate system, then the aligned factors can be merged directly in a shared subspace, dramatically reducing interference. This idea has a rich precedent in the broader merging literature. Entezari et al. (2021) conjectured that all SGD solutions from the same initialisation are linearly mode-connected once permutation symmetries are resolved, known as the permutation conjecture. Git Re-Basin

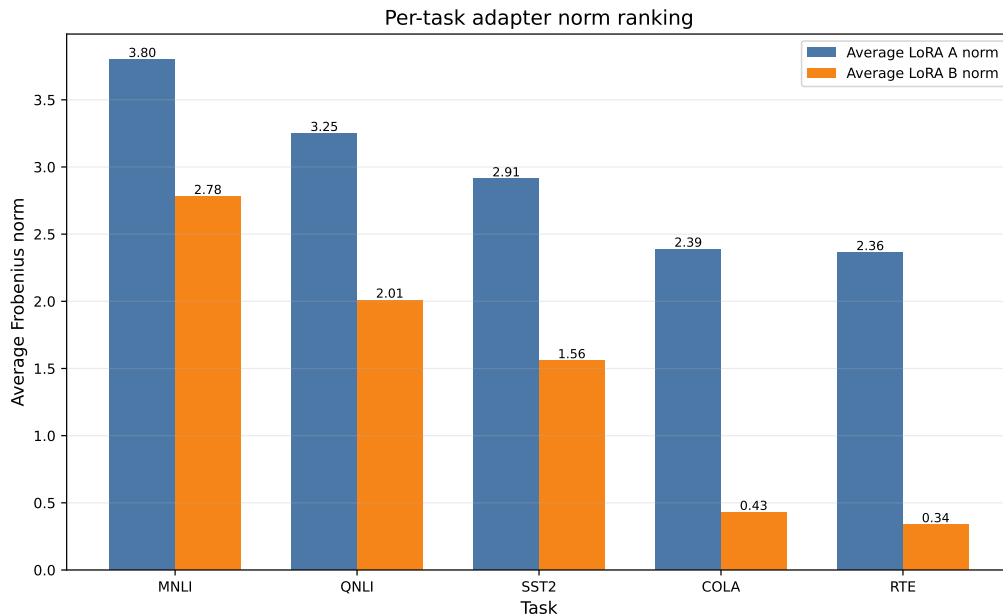


Figure 1.2: Per-task adapter norm ranking for five QLoRA adapters trained on GLUE tasks using Qwen2.5-1.5B. MNLI’s average LoRA B norm (2.78) exceeds RTE’s (0.34) by over $8\times$, illustrating the scale imbalance that causes large-norm adapters to dominate naive merging. This imbalance motivates the equal-contribution property of GPA alignment and the scale-aware enhancements proposed in Chapter 3.

(Ainsworth, Hayase and Srinivasa, 2022) validated this for pairwise merging, using weight and activation matching to permute one model into alignment with a reference before averaging. For LoRA adapters, KnOTS (Stoica et al., 2024) proposed aligning the representation spaces of multiple LoRA models via a single-pass SVD of the concatenated task updates, obtaining a shared left singular basis into which the per-task right singular factors are merged. However, this concatenation approach conflates alignment with merging and, crucially, weights each adapter’s contribution to the shared basis by its Frobenius norm: adapters trained on larger or harder tasks, which tend to develop larger-norm factors, implicitly dominate the consensus. Furthermore, the concatenation requires materialising an $Nr \times d$ matrix, creating an $O(N)$ memory cost in the alignment phase. While this overhead is negligible for the five-adapter setting studied here, it scales linearly with both the number of adapters and the hidden dimension of the base model, becoming significant for larger-scale deployment. When adapters are trained on tasks of unequal difficulty or dataset size, the typical setting in practice, the norm bias systematically distorts the alignment.

This dissertation proposes applying Generalized Procrustes Analysis (GPA) (Gower, 1975) to simultaneously align the A factor matrices of N independently trained LoRA adapters into a shared coordinate system. GPA, originally introduced in morphometrics, finds orthogonal rotation matrices $\{Q_i\}$ that minimise the total sum of squared distances from each rotated factor to an iteratively updated consensus configuration $C = \frac{1}{N} \sum_i Q_i A_i$. The key advantages over prior alignment strategies are threefold. First, GPA is *simultaneous*: all adapters are aligned together, avoiding the reference-model bias inherent in sequential pairwise approaches. Second, GPA is *symmetric*: each adapter contributes equally to the consensus regardless of its norm, which is the correct inductive bias when adapters are drawn from heterogeneous tasks. Third, the alignment is *geometrically principled*: each iteration solves a sequence of orthogonal Procrustes

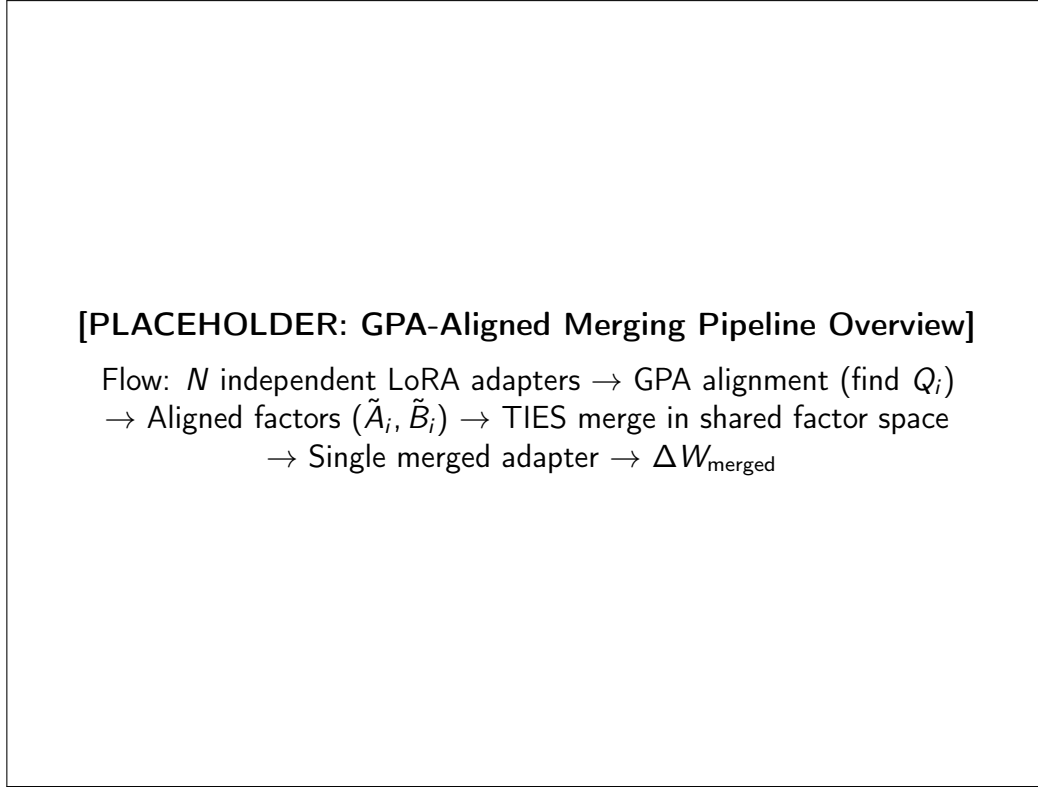


Figure 1.3: Schematic overview of the GPA-aligned merging pipeline. N independently trained LoRA adapters are first aligned into a shared coordinate system via Generalized Procrustes Analysis, which finds per-adapter orthogonal rotations Q_i . The aligned A factors are merged using TIES in the shared factor space, while the counter-rotated B factors are averaged. The merged factors are then reconstructed into a single adapter ΔW_{merged} . The alignment step operates on tiny $r \times r$ SVDs and adds negligible computational overhead.

problems (Schönemann, 1966) in closed form via SVD, with well-understood convergence properties (Ling, 2021). Once aligned, the B factors are counter-rotated ($\tilde{B}_i = B_i Q_i^T$) to preserve functional invariance ($\tilde{B}_i \tilde{A}_i = B_i A_i$ exactly), and any downstream merging strategy (TIES, simple averaging, or otherwise) is then applied in the shared coordinate system. The alignment operates on the tiny $r \times d_{\text{in}}$ matrices directly, with SVD performed on $r \times r$ matrices of dimension as small as 16×16 for rank-16 adapters, maintaining an $O(1)$ alignment-phase memory footprint regardless of the number of adapters. Figure 1.3 provides a schematic overview of the complete GPA-aligned merging pipeline.

The contributions of this dissertation are as follows:

1. A novel application of Generalized Procrustes Analysis to LoRA factor-space alignment for multi-adapter merging, providing simultaneous multi-way alignment without requiring a reference model, training-time constraints, or access to data or gradients.
2. A synthetic validation framework consisting of four experiments that together characterise GPA’s rotation-recovery accuracy, convergence speed, robustness to non-orthogonal perturbations, and ability to preserve dominant shared subspace structure in a realistic LoRA-like generative model.
3. Identification and diagnosis of *scale bias* in factor-space merging pipelines, motivated by the adapter norm analysis reported in Chapter 4, with three targeted scale-aware

enhancements: directional GPA via pre-normalisation of A factors, scale-aware TIES merging on the rescaled aligned factors, and inverse-norm-weighted $\tilde{B}$ averaging.

4. A systematic empirical comparison of GPA+TIES against five baseline merging methods (Task Arithmetic, TIES-Merging, DARE, LR-KnOTS, and unaligned averaging) on a five-task GLUE subset using Qwen2.5-1.5B with QLoRA adapters, with full ablations over the scaling coefficient, LoRA rank, and number of merged adapters.

The remainder of this dissertation is organised as follows. Chapter 2 surveys the relevant literature across model merging, Procrustes analysis in machine learning, and the theoretical properties of the fine-tuning loss landscape that make weight-space alignment feasible. Chapter 3 presents the GPA alignment algorithm, the factor-space merging pipeline, and the scale-aware enhancements in detail. Chapter 4 reports the synthetic validation experiments and the full empirical comparison. Chapter 5 discusses the implications of the results, the limitations of the approach, and directions for future work.

Chapter 2

Related Work

The work presented in this dissertation sits at the intersection of three active research threads: parameter-efficient fine-tuning and the structure of LoRA adapters; the theory and practice of model merging; and the application of Procrustes analysis to representation alignment in machine learning. This chapter surveys each thread in turn, building toward the gap that motivates the present contribution.

2.1 Parameter-Efficient Fine-Tuning and LoRA

Understanding why independently trained LoRA adapters are now abundant, and why their factored structure creates both the alignment problem and the opportunity for geometric solutions, is essential context for the method proposed in Chapter 3. The dominant paradigm for adapting large pre-trained language models to downstream tasks was, for many years, full fine-tuning: updating all model parameters on the target data. This approach achieves strong performance but incurs costs proportional to model size, both in the memory required to store gradients and optimiser states and in the compute required to propagate them. As model scales have grown into the billions and hundreds of billions of parameters (Brown et al., 2020), full fine-tuning has become inaccessible except to well-resourced actors, creating strong demand for methods that confine the update budget while preserving task performance.

Parameter-efficient fine-tuning methods address this by introducing a small number of trainable parameters while keeping the pre-trained weights frozen, thereby preventing catastrophic forgetting (McCloskey and Cohen, 1989) of general capabilities while acquiring task-specific ones. Adapter methods (Houlsby et al., 2019) insert small bottleneck modules between Transformer layers, achieving within 0.4% of full fine-tuning on GLUE while adding only 3.6% additional parameters per task. Prefix-tuning (Li and Liang, 2021) and prompt tuning (Lester, Al-Rfou and Constant, 2021) prepend learnable continuous vectors to the input or intermediate activations, enabling adaptation with even fewer parameters. These methods established the PEFT paradigm but typically introduce inference latency through added modules or extended input sequences.

Low-Rank Adaptation (Hu et al., 2022) addresses the latency problem by reparameterising the weight update of each Transformer layer as $\Delta W = BA$, where $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ with rank $r \ll \min(d_{\text{in}}, d_{\text{out}})$. Only A and B are trained; the pre-trained weight W_0 is frozen. At inference time the update ΔW can be absorbed into W_0 , adding zero latency compared to

the unmodified model. The empirical success of LoRA rests on a deeper theoretical property: as Aghajanyan, Gupta and Zettlemoyer (2021) demonstrated, pre-trained language models have a very low intrinsic dimensionality for fine-tuning. Optimising in a randomly projected subspace of only a few hundred dimensions recovers most of the performance of full fine-tuning on a range of NLP tasks, and better pre-training yields *lower* intrinsic dimension. This finding, building on the earlier work of Li et al. (2018) on the intrinsic dimension of objective landscapes, provides the theoretical justification for the low-rank constraint: fine-tuning updates genuinely live in low-dimensional subspaces, not merely as an approximation but as a reflection of the structure of pre-trained representations.

QLoRA (Dettmers et al., 2023) extended LoRA to quantised base models, combining 4-bit NF4 quantisation of frozen weights with 16-bit LoRA adapters to enable the fine-tuning of 65-billion-parameter models on a single 48 GB GPU. By dramatically lowering the hardware barrier to LoRA fine-tuning, QLoRA has substantially expanded the practical population of independently trained LoRA adapters and therefore increased the need for ensemble alignment methods. AdaLoRA (Zhang et al., 2023b) extended the LoRA framework by parameterising updates via SVD decomposition and adaptively allocating rank budgets across weight matrices based on importance scores, making the singular structure of the update matrices explicit.

The factored structure of LoRA creates both the problem and the opportunity addressed by this dissertation. The update $\Delta W = BA$ is not uniquely decomposed: for any invertible $r \times r$ matrix M , $(BM)(M^{-1}A)$ represents the same update. In practice, each independent fine-tuning run chooses a particular rotation of the shared latent subspace, producing A matrices that span geometrically equivalent but coordinate-wise distinct subspaces. This rotational ambiguity is the geometric root of the interference observed when LoRA adapters trained on different tasks are merged: it is not merely statistical noise but a systematic misalignment of the bases in which task-specific features are encoded. Exposing and resolving this misalignment is the central purpose of the GPA framework proposed here.

2.2 Model Merging

This section traces the evolution of merging methods from simple averaging through conflict resolution to geometric alignment, establishing that existing approaches leave a gap in simultaneous multi-way alignment of LoRA factor matrices. Model merging refers to the family of techniques that combine independently trained models by operating directly in weight space, producing a single model that incorporates capabilities from multiple sources without any further training. The field has developed rapidly, producing a progression of increasingly principled methods for managing the destructive interference that naïve averaging incurs.

2.2.1 Weight Averaging and Task Arithmetic

The foundational observation underlying model merging is that models fine-tuned from a shared pre-trained initialisation tend to occupy the same region of parameter space. Model Soups (Wortsman et al., 2022) demonstrated that averaging the weights of multiple models fine-tuned with different hyperconfigurations from the same checkpoint consistently improves accuracy and robustness without increasing inference cost, and established the key intuition that fine-tuned models from a shared initialisation lie within a single low-error basin.

Task Arithmetic (Ilharco et al., 2022) formalised this observation by introducing the concept

of a *task vector*: the signed difference $\tau = \theta_{\text{ft}} - \theta_{\text{pre}}$ between the fine-tuned and pre-trained parameters. Task Arithmetic showed that these vectors can be composed by arithmetic: adding multiple task vectors to the pre-trained model produces a multi-task model, while negating a task vector removes the corresponding capability. The method also introduced the scaling coefficient λ that modulates the magnitude of the applied update, providing a simple hyperparameter for balancing task contributions. Task Arithmetic constitutes the simplest and most broadly applicable merging baseline and the conceptual framework within which most subsequent methods operate. Zhang et al. (2023a) extended this arithmetic framework to parameter-efficient modules including LoRA, defining addition and negation operators for composing PEFT modules for multi-tasking and unlearning.

2.2.2 Conflict Resolution Methods

While Task Arithmetic demonstrated the viability of weight-space composition, it also exposed systematic failure modes when the merged task vectors conflict. TIES-Merging (Yadav et al., 2023) provided a detailed analysis of the interference problem, identifying three distinct sources: redundant parameters with near-zero change that contribute noise; conflicting signs between models that cause partial cancellation; and magnitude imbalance that allows large-norm adapters to dominate. TIES addresses these through a three-step procedure: trim parameters below a magnitude threshold; elect a consensus sign for each parameter via majority vote weighted by magnitude; and merge only the parameters whose sign agrees with the elected direction. Extensive experiments across vision and language benchmarks demonstrated consistent improvements over Task Arithmetic, particularly as the number of merged models grows.

DARE (Yu et al., 2024) proposed a complementary approach motivated by the observation that most fine-tuning updates are redundant and can be randomly pruned before merging without significant performance loss. By applying Bernoulli sparsification to the task vectors at high drop rates (up to 99% for large models), DARE reduces the total magnitude of interfering updates, and the resulting sparse vectors can be applied on top of any downstream merging method. Both TIES and DARE treat interference as arising from conflicting scalar parameter values and address it through sparsification and sign resolution. They do not consider the geometric structure of the update matrices or the possibility that interference arises from misaligned low-rank subspaces, a limitation that becomes critical in the LoRA setting where the factored $B \times A$ structure makes subspace rotations explicit. Fisher Merging (Matena and Raffel, 2022) took a more principled statistical approach, computing diagonal Fisher information matrices as per-parameter importance scores to weight the merging contribution of each model. RegMean (Jin et al., 2022) computed optimal merging weights by minimising prediction discrepancies using inner-product matrices of layer activations, exploiting structural information about how each layer processes inputs. While both methods provide principled weighting schemes, neither addresses the geometric alignment of the subspaces in which the updates operate. Figure 2.1 situates these approaches within a taxonomy of merging methods, distinguishing statistical conflict resolution from geometric alignment and pairwise from simultaneous multi-way methods.

2.2.3 Alignment-Before-Merging

A distinct line of work recognises that the fundamental obstacle to effective model merging is not statistical noise but geometric misalignment arising from the symmetries of neural network

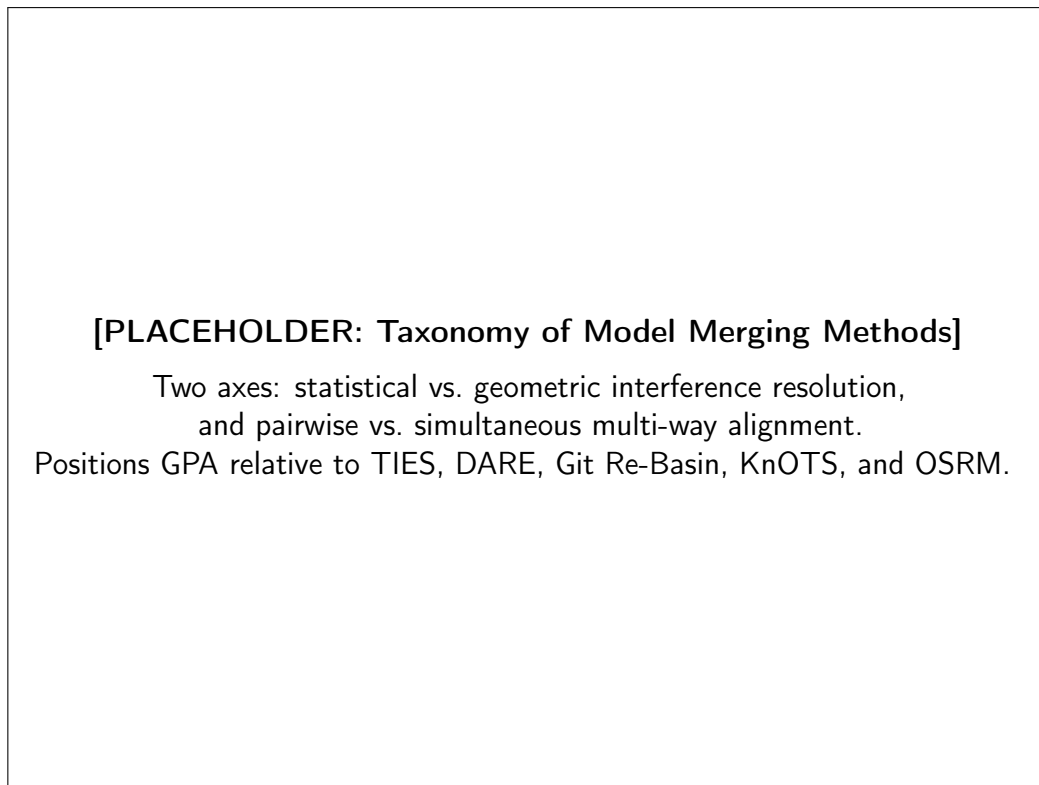


Figure 2.1: Taxonomy of model merging methods. Existing approaches occupy three quadrants: statistical conflict resolution without alignment (TIES, DARE, Fisher Merging), geometric pairwise alignment (Git Re-Basin), and single-pass geometric alignment (KnOTS). GPA occupies the previously empty quadrant of simultaneous multi-way geometric alignment, treating all adapters symmetrically without a reference model. OSRM enforces orthogonality at training time and is therefore complementary to post-hoc methods.

parameterisations. The permutation conjecture, formulated by Entezari et al. (2021), holds that all SGD solutions from the same random initialisation are linearly mode-connected once permutation symmetries of the hidden units are accounted for. This conjecture implies that the apparent barriers to linear interpolation between independently trained models are artefacts of symmetry rather than genuine loss barriers: resolve the permutation symmetry and the models lie on a connected low-loss path. Git Re-Basin (Ainsworth, Hayase and Srinivasa, 2022) provided strong empirical validation of this conjecture, introducing three algorithms (weight matching, activation matching, and straight-through estimation) that permute the hidden units of one model into correspondence with a reference before merging. The resulting aligned models can be averaged with dramatically reduced loss barriers. Git Re-Basin is the most direct precursor to the GPA framework proposed here: it demonstrates that alignment prior to merging is essential for full-model weight averaging. However, Git Re-Basin operates pairwise and aligns to a fixed reference, introducing an asymmetry that GPA’s simultaneous, reference-free formulation avoids.

2.2.4 LoRA-Specific Merging Methods

Merging independently trained LoRA adapters poses challenges beyond those of full-model merging, because the factored $\Delta W = BA$ structure introduces an additional rotational symmetry that is absent in full-rank weight matrices. LoRAHub (Huang et al., 2023) uses

gradient-free optimisation to learn scalar coefficients for linearly combining LoRA modules trained on diverse tasks, enabling cross-task generalisation from a small number of target examples. The method implicitly assumes that LoRA modules are already in compatible coordinate systems and does not attempt to resolve subspace misalignment. ZipLoRA (Shah et al., 2024) directly observed that independently trained LoRA modules for diffusion models exhibit destructive interference when columns of their A matrices have high cosine similarity across adapters, and proposed learning column-wise disjoint merger coefficients to minimise this. This work provides compelling empirical evidence that rotated subspaces are a genuine and damaging phenomenon in LoRA merging, though ZipLoRA’s approach of sparsity-based disentanglement is distinct from the explicit geometric alignment proposed here.

KnOTS (Stoica et al., 2024) is the most closely related prior work. It proposed aligning the representation spaces of multiple LoRA models by concatenating their task updates horizontally and applying a single SVD, obtaining a shared left singular basis U into which the per-task right singular factors $V^{(i)}$ are merged. KnOTS demonstrated consistent improvements over unaligned merging on vision benchmarks and established alignment of LoRA factors as a viable strategy. However, the concatenation approach has two fundamental limitations. First, because the SVD of the stacked matrix $[\Delta W^{(1)} \ \Delta W^{(2)} \ \dots \ \Delta W^{(N)}]$ is least-squares optimal over the concatenation as a whole, adapters with larger Frobenius norms contribute disproportionately to the shared basis U . In the heterogeneous-task setting typical of GLUE, where adapters trained on large datasets such as MNLI have systematically larger norms than those trained on small datasets such as RTE, this norm bias distorts the consensus. Second, KnOTS conflates the alignment step with the merging step, providing no explicit per-adapter rotation matrices that could be inspected as diagnostics or used independently.

OSRM (Zhang and Zhou, 2025) took a fundamentally different approach, constraining the LoRA update subspaces to be orthogonal between tasks *at training time*, thereby preventing interference from arising in the first place. This validates the core hypothesis that subspace interference is the dominant failure mode of LoRA merging, but requires modifying the fine-tuning procedure and cannot be applied post-hoc to already-trained adapters. Isotropic Merging (Marczak et al., 2025) addressed the related problem of scale imbalance by flattening the singular value spectrum of task matrices before merging, so that no individual direction disproportionately dominates the merged update. This manipulation of the *spectrum* is complementary to the manipulation of the *basis* performed by GPA, and the two strategies could in principle be combined.

No existing method performs simultaneous multi-way geometric alignment of LoRA factor matrices while treating all adapters symmetrically and producing explicit per-adapter rotation matrices that can be applied independently of the choice of downstream merging strategy. This is the gap the present work fills.

2.3 Procrustes Analysis and Representation Alignment

The mathematical framework underlying the proposed method is classical Procrustes analysis, which provides a family of problems concerned with finding optimal transformations between configurations of points. This section introduces the classical and generalised forms of Procrustes analysis that provide the algorithmic foundation for the proposed method, and reviews established applications to neural representation alignment that provide methodological precedent for the LoRA setting.

2.3.1 Classical and Generalised Procrustes Analysis

The orthogonal Procrustes problem, in its modern formulation, asks for the orthogonal matrix $T^* = \arg \min_T \|AT - B\|_F$ that best aligns a matrix A to a target B . Schönemann (1966) provided the closed-form solution: compute the SVD $B^\top A = U\Sigma V^\top$ and set $T^* = UV^\top$. This solution underlies every iteration of the GPA algorithm: at each step, each adapter’s rotation is updated by solving a pairwise Procrustes problem between the current rotated factor and the consensus matrix.

Gower (1975) introduced Generalised Procrustes Analysis as the multi-way extension: given N configurations $\{A_i\}$, find orthogonal rotations $\{Q_i\}$ that minimise the total sum of squared distances to the consensus $C = \frac{1}{N} \sum_i Q_i A_i$. The alternating optimisation, fix the consensus and update all rotations in closed form via Schönemann’s solution, then update the consensus as the mean of the rotated configurations, is guaranteed to monotonically decrease the objective and converges in practice in a small number of iterations. Critically, GPA has no reference model: all configurations are treated on equal terms, avoiding the asymmetry that would result from aligning all adapters to one arbitrarily chosen reference. Ling (2021) established near-optimal convergence bounds for the generalised orthogonal Procrustes problem via the generalised power method, providing theoretical guarantees on the rate at which the algorithm approaches a stationary point. Pizarro and Bartoli (2011) analysed the global optimisation landscape of GPA, providing conditions under which the alternating algorithm avoids poor local minima.

2.3.2 Procrustes Alignment in Machine Learning

The observation that independently trained models can be aligned via Procrustes transformations has a substantial history in the neural representation literature, providing direct precedent for applying GPA to LoRA factor matrices. Conneau et al. (2017) demonstrated that independently trained monolingual word embedding spaces can be aligned across languages using adversarial training followed by iterative Procrustes refinement, enabling bilingual dictionary induction without parallel data. The core insight, that independently trained embedding spaces share similar geometric structure but exist in coordinate systems related by an orthogonal transformation, is directly analogous to the situation for independently trained LoRA adapters and provides strong empirical precedent for the GPA approach.

Williams et al. (2021) formalised Procrustes alignment as a metric on neural representations, showing that it provides a well-motivated distance measure satisfying the triangle inequality and grounding the treatment of GPA alignment as a principled geometric operation rather than an ad-hoc heuristic. More recently, Achara et al. (2026) extended Procrustes-based representation alignment to the multi-way setting, studying the simultaneous alignment of representations learned across multiple modalities and architectures. This work provides the most direct parallel in the representation alignment literature for the multi-way LoRA alignment problem considered here, demonstrating that GPA-style simultaneous alignment yields stronger results than sequential pairwise alignment when multiple representation spaces must be reconciled.

2.3.3 Loss Landscape Theory and the Feasibility of Alignment

The preceding sections establish that Procrustes alignment is mathematically well-founded and has proven effective for neural representations. A natural question remains: why should we expect LoRA adapters fine-tuned from the same base model to share enough subspace

structure for GPA alignment to recover something meaningful? The answer lies in the geometry of the fine-tuning loss landscape.

Garipov et al. (2018) discovered that independently trained neural networks are connected by simple curves of near-constant loss in weight space, a phenomenon termed mode connectivity. Frankle et al. (2020) strengthened this to *linear* mode connectivity for models that share a portion of their optimisation trajectory, demonstrating that models fine-tuned from a common checkpoint can often be linearly interpolated without loss degradation. Neyshabur, Sedghi and Zhang (2020) provided direct empirical evidence that models fine-tuned from the same pre-trained checkpoint remain in the same loss basin, while models from different initialisations exhibit a sharp interpolation barrier. Together, these results imply that the fine-tuning loss landscape is globally structured by the pre-trained initialisation: adapters derived from the same base model share a geometric neighbourhood in parameter space, and linear interpolation between them is approximately loss-preserving.

The intrinsic dimensionality results of Li et al. (2018) and Aghajanyan, Gupta and Zettlemoyer (2021) add a further structural observation: not only do fine-tuned models share a loss basin, but the updates that carry them from the pre-trained initialisation to the fine-tuned solution lie in low-dimensional subspaces. Aghajanyan, Gupta and Zettlemoyer (2021) showed that optimising only a few hundred randomly projected parameters can recover most of the performance of full fine-tuning for well-pre-trained models, and that larger and better pre-trained models have *lower* intrinsic dimension for fine-tuning. This result provides the theoretical justification for LoRA’s low-rank constraint and, by extension, for the GPA alignment strategy: if fine-tuning updates are intrinsically low-rank, then the subspaces spanned by different adapters’ A matrices should overlap substantially, making geometric alignment both meaningful (there is a genuine shared structure to recover) and feasible (the rotational misalignment can be resolved without access to data).

Centered Kernel Alignment (CKA) (Kornblith et al., 2019) provides an empirical tool for measuring the similarity between neural network representations that is invariant to orthogonal transformations and isotropic scaling. CKA has been used extensively to establish that different architectures, training procedures, and initialisations converge to qualitatively similar representational structures, particularly in the intermediate layers of deep networks. The Platonic Representation Hypothesis (Huh et al., 2024) generalises this observation, arguing that representations across models, architectures, and even modalities are converging toward a shared statistical structure reflecting the underlying reality on which they are trained. If this hypothesis holds even approximately for LoRA adapters derived from the same base model, it provides deep theoretical motivation for the GPA alignment strategy: the shared structure being aligned is not arbitrary but reflects genuine task-invariant features learned during pre-training. In this dissertation, CKA is used as an evaluation metric to quantify the degree of representational alignment achieved by GPA before and after the alignment step.

The collective implication of these findings is that LoRA adapters fine-tuned from the same base model should share substantial subspace structure. The loss landscape results explain why weight-space operations are viable at all; the intrinsic dimensionality results explain why the low-rank factor space is the appropriate domain for alignment; and the representational similarity results explain why GPA can be expected to recover a meaningful shared structure rather than imposing an arbitrary geometric transformation on unrelated subspaces. Against this theoretical backdrop, the absence of a principled multi-way alignment method for LoRA factor matrices represents a clear gap in the existing literature, a gap that the present work

addresses.

Chapter 3

Methodology

The previous chapter closed with a gap: no existing method performs simultaneous multi-way geometric alignment of LoRA factor matrices while treating all adapters symmetrically and producing explicit, interpretable per-adapter rotations. This chapter sets out the experimental programme by which we address that gap. Five claims advanced in Chapter 1 structure everything that follows: (C1) GPA is a validated alignment primitive for low-rank factor spaces; (C2) alignment quality and merge quality are separable problems, so good alignment does not by itself imply a good merged model; (C3) scale imbalance, not rotational misalignment, is the binding constraint on factor-space LoRA merging; (C4) within the compact, rank-preserving family of factor-space merging methods, the enhanced GPA+TIES pipeline is competitive with or modestly superior to its most direct baselines; and (C5) the CoLA task exposes a structural limit of compact factor-space merging that no current method resolves. The chapter first fixes notation (Section 3.1), then describes every method under test (Section 3.2), then states the twelve experiments whose results Chapter 4 reports (Section 3.3), and finally documents the evaluation protocol (Section 3.4).

3.1 Problem Setting and Notation

Let $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ denote a frozen weight matrix from a pre-trained base model. A LoRA adapter (Hu et al., 2022) parameterises a task-specific update by a pair of low-rank factors (A_i, B_i) with $A_i \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B_i \in \mathbb{R}^{d_{\text{out}} \times r}$, and applies the adapted weight $W_i = W_0 + B_i A_i$. We consider the setting in which N adapters $\{(A_i, B_i)\}_{i=1}^N$ have been independently fine-tuned on N different downstream tasks from the same shared base W_0 at a common rank r . The objective is to construct a single merged adapter $(A_{\text{merged}}, B_{\text{merged}})$, also of rank r , without access to any training data or gradients.

The central algebraic fact the chapter rests on is that the factored representation $B_i A_i$ is invariant under any orthogonal transformation $Q_i \in \mathbb{R}^{r \times r}$ with $Q_i^\top Q_i = I$:

$$B_i Q_i^\top Q_i A_i = B_i A_i. \quad (3.1)$$

This functional-invariance identity licences every alignment-before-merging method discussed in this chapter: rotating the A factor and counter-rotating the B factor leaves the adapter’s action on any input exactly unchanged. In contrast, any operation that averages or sparsifies the A factors pointwise is *not* invariant under this rotational freedom. This asymmetry is the

geometric origin of the destructive interference previewed in Figure 1.1: an averaging operation applied to factors whose rows encode the same features in different coordinate systems cancels rather than reinforces. The methods under test in Section 3.2 differ primarily in whether, and how, they exploit the rotational freedom of Equation 3.1 before merging.

3.2 Methods Under Test

This section describes each merging method evaluated in Chapter 4. Baselines are covered compactly because they were positioned in Chapter 2; the proposed method and its scale-aware enhancements receive the fuller treatment.

3.2.1 Simple Averaging

The elementary baseline averages the factors pointwise, $A_{\text{merged}} = \frac{1}{N} \sum_i A_i$ and $B_{\text{merged}} = \frac{1}{N} \sum_i B_i$, with no alignment and no conflict resolution. This is the null experiment: if neither alignment nor statistical conflict resolution contributed anything, this method would be indistinguishable from the rest. It is included to calibrate the magnitude of every other method's improvement.

3.2.2 Task Arithmetic

Task Arithmetic (Ilharco et al., 2022) merges models in the space of full-rank task deltas: $\Delta W_{\text{merged}} = \lambda \sum_i B_i A_i$, with no alignment and no sparsification. The scaling coefficient λ controls the magnitude of the applied update and is swept as part of the evaluation. A property that recurs throughout the analysis is that the summed delta $\sum_i B_i A_i$ has rank at most Nr : for the five-adapter, rank-16 setting studied here, the resulting update is rank-80. Task Arithmetic is therefore *not* a rank-preserving baseline. Any comparison with factor-space methods is read as a rank-versus-rank comparison rather than a pure merging-quality comparison, a point flagged in Table 3.1 and recalled in the ablations.

3.2.3 TIES-Merging

TIES (Yadav et al., 2023) is a three-step statistical conflict-resolution procedure applied to the task deltas $\Delta W_i = B_i A_i$. First, each delta is *trimmed* by keeping only the top $k\%$ of parameters by absolute magnitude. Second, at each parameter position a consensus sign is *elected* by magnitude-weighted majority vote across the N trimmed deltas. Third, the merged value at each position is the average of only those contributions whose sign agrees with the elected direction. In this dissertation, TIES is applied in two roles: as a baseline on the unaligned deltas, and as the merging step inside the GPA+TIES pipeline on the aligned A factors. A methodological observation that bears on the interpretation of results in Chapter 4: at the trim setting $k = 20\%$ used below, with $N = 5$ adapters each contributing a nonzero value at a different subset of positions, the disjoint-merge step repopulates nearly every position of the output, so the final merged adapter is effectively dense, with a nonzero fraction very close to one. The benefit of TIES in this regime derives almost entirely from sign election, not from sparsity in the merged representation.

3.2.4 DARE + TIES

DARE (Yu et al., 2024) applies Bernoulli sparsification to the task deltas prior to merging: each parameter is independently dropped with probability p and the survivors are rescaled by $1/(1 - p)$ to preserve the expected magnitude. The sparsified deltas are then passed through TIES. DARE is included as the dominant delta-space sparsification baseline and to test whether stochastic pruning provides regularisation beyond the deterministic trim operation of TIES alone.

3.2.5 LR-KnOTS + TIES

LR-KnOTS is the LoRA adaptation of KnOTS (Stoica et al., 2024) used as the factor-space alignment baseline. The per-adapter A_i matrices are concatenated horizontally into a single $r \times Nd_{\text{in}}$ matrix, a single SVD is taken of this concatenation, and each adapter is projected onto the shared left singular basis; the projected factors are merged with TIES in this shared basis and the output is reconstructed. LR-KnOTS is the closest direct competitor to the proposed method: both perform alignment followed by factor-space merging, and both are rank-preserving. As discussed in Chapter 2, the two methods differ in that LR-KnOTS is single-pass and concatenation-based, weighting each adapter’s contribution to the shared basis by its Frobenius norm, whereas the proposed method is iterative and symmetric. The head-to-head comparison between GPA+TIES and LR-KnOTS+TIES is therefore the primary empirical test of claim C4.

3.2.6 GPA+TIES (Proposed Method)

The proposed method performs Generalized Procrustes Analysis (Gower, 1975) on the LoRA A factors, followed by TIES merging in the aligned factor space. GPA finds per-adapter orthogonal rotations $\{Q_i\}_{i=1}^N$ and a consensus matrix $C \in \mathbb{R}^{r \times d_{\text{in}}}$ that solve

$$\min_{\{Q_i\}, C} \sum_{i=1}^N \|Q_i A_i - C\|_F^2 \quad \text{subject to} \quad Q_i^\top Q_i = I \quad \forall i. \quad (3.2)$$

The algorithm alternates between two closed-form updates. With the consensus C held fixed, each rotation is updated by solving an orthogonal Procrustes problem (Schönemann, 1966): compute the SVD $CA_i^\top = U_i \Sigma_i V_i^\top$ and set $Q_i = U_i V_i^\top$. With the rotations held fixed, the consensus is updated as the mean $C = \frac{1}{N} \sum_i Q_i A_i$. Each update is guaranteed to not increase the objective, so the sequence of objective values converges; empirically the iteration converges in a small number of passes, as tested in Experiment 2. Ling (2021) provide near-optimal convergence bounds for this algorithm and Pizarro and Bartoli (2011) analyse conditions on the global-optimum landscape. The SVDs performed at each iteration are of $r \times r$ matrices, so the per-iteration cost is independent of d_{in} .

Once GPA has produced the rotations $\{Q_i\}$, the aligned factors $\tilde{A}_i = Q_i A_i$ and counter-rotated factors $\tilde{B}_i = B_i Q_i^\top$ preserve the adapter’s function exactly by Equation 3.1. The aligned $\tilde{A}_i$ are merged using TIES to produce A_{merged} ; the counter-rotated $\tilde{B}_i$ are averaged to produce B_{merged} . The merged adapter is applied as $W_0 + \lambda B_{\text{merged}} A_{\text{merged}}$, with the scaling coefficient λ treated as a hyperparameter on the same footing as for Task Arithmetic and TIES. The full pipeline is applied *independently at each of the 196 LoRA modules* of the Qwen2.5-1.5B base model (28 transformer layers $\times$ 7 LoRA-equipped module types). The structural analysis of

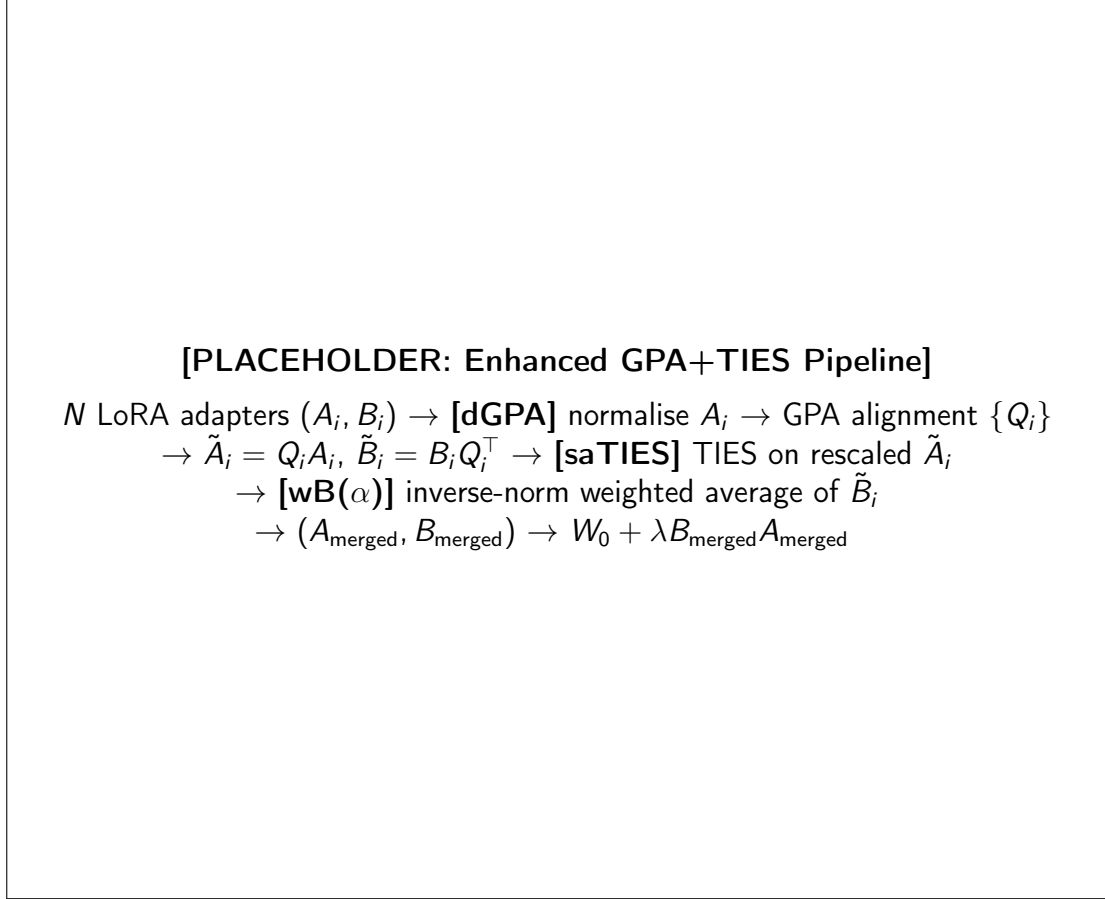


Figure 3.1: Schematic overview of the enhanced GPA+TIES pipeline, with the three scale-aware modifications of Section 3.2.7 annotated. The baseline GPA+TIES corresponds to disabling all three scale-aware steps: running GPA on the unnormalised A_i , merging with standard TIES, and averaging $\tilde{B}_i$ uniformly. The pipeline is applied independently at each of the 196 LoRA modules of Qwen2.5-1.5B; per-module rotations Q_i are retained as diagnostic outputs, distinguishing the method from baselines that conflate alignment with merging.

Experiment 5 shows that norm and directional heterogeneity are organised at the module level rather than globally, so a single model-wide rotation would not be able to reconcile them. The full pipeline is illustrated schematically in Figure 3.1.

3.2.7 Enhanced GPA+TIES Variants

The baseline GPA+TIES pipeline treats all adapters symmetrically at the alignment step but inherits TIES’ and simple averaging’s norm-sensitivity at the merging and B steps. Three scale-aware enhancements, each motivated by a failure mode diagnosed in the adapter structural analysis of Experiment 5, are introduced and can be combined freely.

Directional GPA (dGPA). In the rotation update $Q_i = U_i V_i^\top$ derived from $CA_i^\top = U_i \Sigma_i V_i^\top$, the matrix A_i enters linearly, so an adapter with Frobenius norm $k\|A_j\|_F$ contributes k times as much to the rotation geometry as A_j and k^2 times as much to the consensus objective value. Given the order-of-magnitude norm spread observed in Figure 1.2, the standard GPA consensus is pulled toward the largest-norm adapter. Directional GPA normalises each A_i to unit Frobenius norm prior to alignment, runs GPA on the normalised factors to obtain

direction-only rotations, and applies those rotations to the *original* unnormalised matrices. The rotations therefore depend only on the directional structure of each adapter, giving each adapter an equal geometric vote in the consensus.

Scale-aware TIES (saTIES). The TIES trim and sign-election steps use absolute magnitudes, so even after directional alignment the large-norm adapters dominate the elected consensus direction at each parameter position. Scale-aware TIES first rescales each aligned $\tilde{A}_i$ to unit Frobenius norm, performs TIES on the normalised factors, and then rescales the merged output to the average of the original aligned norms. This decouples the sign-election vote from adapter scale while preserving the overall output magnitude.

Inverse-norm B weighting ($wB(\alpha)$). Under simple averaging of the counter-rotated $\tilde{B}_i$, adapters with large B norms dominate the output in proportion to their norm. The inverse-norm B weighting replaces the uniform average by $B_{\text{merged}} = \sum_i w_i \tilde{B}_i$ with $w_i \propto \|\tilde{B}_i\|_F^{-\alpha}$, normalised so that $\sum_i w_i = 1$. The exponent α interpolates between uniform averaging at $\alpha = 0$ and strong compensation toward the smallest-norm adapter at larger α .

The three enhancements are composable, giving rise to the five variants evaluated in Chapter 4: GPA+TIES, dGPA+TIES, dGPA+saTIES, dGPA+saTIES+wB(0.5), and dGPA+saTIES+wB(1.0). This is the set used in the enhancement ablation (Experiment 9). The geometric effect of the directional-alignment step is illustrated in Figure 3.2: under standard GPA, a large-norm adapter pulls the consensus toward its own direction; under directional GPA, the consensus is determined by adapter directions alone and each adapter contributes equally regardless of its norm.

3.2.8 Qualitative Method Comparison

Table 3.1 summarises the methodological properties of the non-trivial methods under test, enabling their results in Chapter 4 to be read in context.

Two properties carry particular weight for reading the subsequent results. First, Task Arithmetic is the only method in the comparison that does not preserve rank, so any comparison against it is properly read as a rank-versus-rank comparison rather than a pure merging-quality comparison. Second, GPA+TIES and LR-KnOTS+TIES are the only rank-preserving methods with an explicit alignment stage, which makes their head-to-head comparison the core test of claim C4.

3.3 Experimental Programme

The experimental programme consists of twelve named experiments grouped into four clusters: synthetic validation of the GPA primitive; real-adapter structural analysis; the main empirical comparison; and targeted ablations. Each experiment is stated in the same compact template of *Aim*, *Protocol*, *Supports*, and *Reads as*, so that every experiment is auditable against a specific claim from Chapter 1.

3.3.1 Synthetic Validation of the GPA Primitive

Synthetic experiments are the natural starting point because on real trained LoRA adapters there is no ground-truth rotation to recover: any assessment of alignment quality on real data

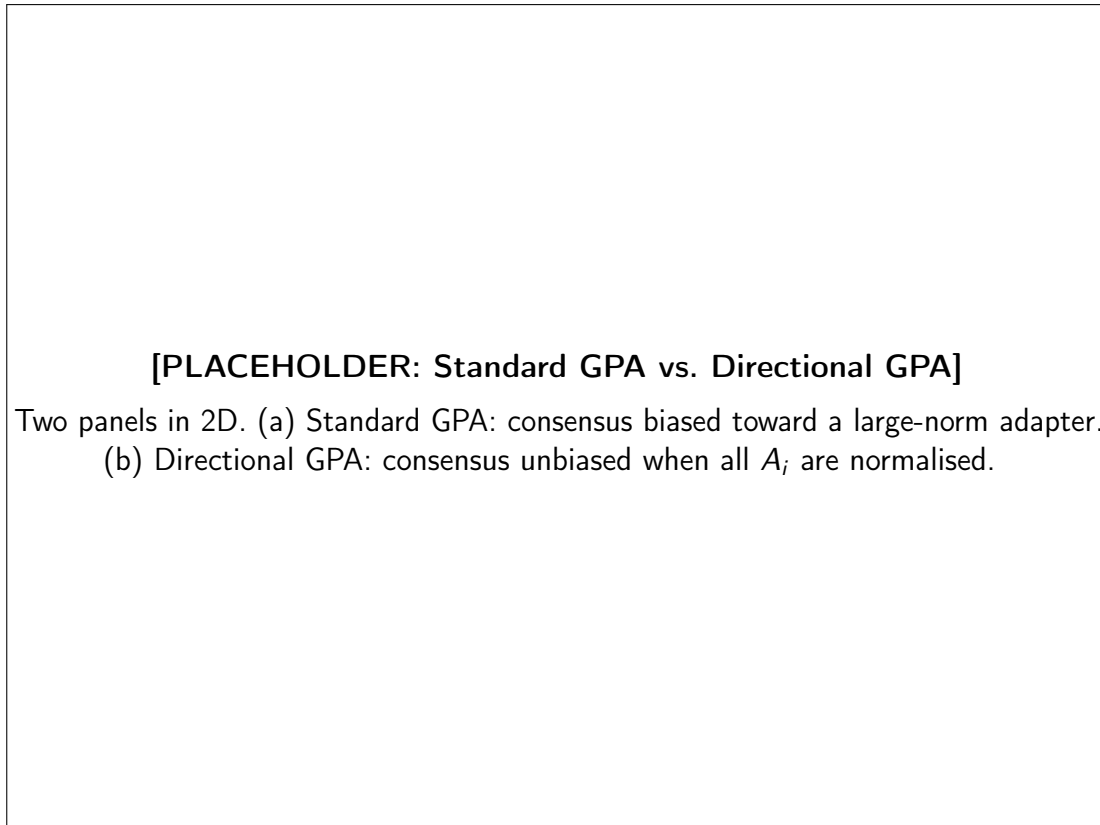


Figure 3.2: Geometric intuition for directional GPA (dGPA). (a) In standard GPA, each A_i enters the rotation update linearly, so adapters with larger Frobenius norm exert proportionally stronger geometric pull on the consensus C , producing a biased centre. (b) Normalising each A_i to unit Frobenius norm before alignment removes the norm dependence from the consensus while preserving directional structure; the recovered rotations are then applied to the original unnormalised matrices for merging, so the norm information is not lost, only removed from the *alignment* objective.

is therefore relative, never absolute. Only a synthetic setting, in which the true rotations are known by construction, can establish the correctness of the GPA primitive itself. The four experiments below isolate claim C1 (the alignment primitive works) from the merge-quality questions that arise on real data (C2–C5).

Experiment 1: Ground-truth rotation recovery. *Aim.* Establish that GPA recovers known orthogonal rotations from noisy observations, the cleanest possible test of correctness. *Protocol.* Sample a reference matrix $A^* \in \mathbb{R}^{16 \times 1536}$ with i.i.d. Gaussian entries. Generate N observations $A_i = Q_i A^* + \sigma E_i$ with random orthogonal Q_i drawn from the Haar measure and i.i.d. Gaussian noise E_i . Sweep the noise scale $\sigma \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5\}$, the number of observations $N \in \{3, 5, 10\}$, and the rank $r \in \{4, 8, 16, 32\}$, with 100 trials per cell. Report rotation-recovery error $\frac{1}{N} \sum_i \|\hat{Q}_i - Q_i\|_F^2$, relative consensus error $\|C - A^*\|_F / \|A^*\|_F$, and alignment residual. *Supports. C1.* *Reads as.* Small rotation-recovery error across the (N, r) grid that degrades only gracefully with σ establishes GPA as a correct alignment primitive in the regime used by all subsequent experiments.

[PLACEHOLDER: Qualitative Method Comparison Table]

7 rows (methodological properties) $\times$ 5 columns (TA, TIES, DARE+TIES, LR-KnOTS+TIES, GPA+TIES). Properties: alignment stage, iterative refinement, convergence guarantee, equal contribution, interpretable rotations, rank preservation, per-module alignment cost.

Table 3.1: Qualitative comparison of methodological properties across the merging methods under test. Task Arithmetic is the only non-rank-preserving method and is therefore read as a rank/capacity comparator rather than a like-for-like merging baseline. GPA+TIES and LR-KnOTS+TIES are the only rank-preserving methods that include an alignment stage; the direct comparison between them is the primary empirical test of claim C4. Only GPA+TIES produces per-adapter rotation matrices that can be inspected as diagnostics independently of the downstream merging step.

Experiment 2: Convergence. *Aim.* Establish that GPA converges quickly enough to be used as a drop-in primitive within a merging pipeline rather than as a separate optimisation phase. *Protocol.* Fix $\sigma = 0.1$; sweep $(N, r) \in \{3, 5, 10, 20\} \times \{8, 16, 32\}$; record the objective residual per iteration and the wall-clock time. *Supports.* C1 (the computational-feasibility component), and the cost claims made in Table 3.1. *Reads as.* Convergence within approximately ten iterations across the grid would make GPA as cheap as simple weight averaging for the purpose of the downstream pipeline.

Experiment 3: Robustness to non-orthogonal perturbations. *Aim.* Test whether GPA degrades gracefully when its orthogonality assumption is violated. Real LoRA adapters are emphatically not exact rotations of a single shared matrix, so this experiment probes the regime in which GPA is actually deployed downstream. *Protocol.* Generate $A_i = (Q_i + \delta S_i)A^* + \sigma E_i$ with random S_i of unit Frobenius norm and $\delta \in \{0, 0.01, 0.05, 0.1, 0.2\}$; sweep (N, r) as in Experiment 1. *Supports.* C1. *Reads as.* Sub-linear, non-catastrophic growth of the alignment residual across δ would validate GPA beyond its idealised assumptions; a step-change around $\delta = 0.1$ would undermine the project’s core premise.

Experiment 4: Structured LoRA-like ground truth. *Aim.* The most informative of the synthetic tests: does GPA preserve a dominant shared subspace when observations contain

both shared structure and task-specific directions? This directly mimics the intended use on real LoRA adapters, which share base-model-induced structure and differ by task-specific perturbations. *Protocol.* Generate A^* with a geometrically decaying singular-value spectrum, mimicking trained LoRA spectra. Add task-specific rank-one perturbations P_i of controlled strength, apply random rotations Q_i , and add noise. Measure the subspace overlap between $\text{span}(C)$ and $\text{span}(A^*)$ via principal angles; sweep perturbation strength. *Supports.* **C1**, and serves as the bridge from synthetic to real. *Reads as.* High subspace overlap even at substantial perturbation strength licences the application of GPA to real adapters, where task-specific deviations from a shared structure are unavoidable.

3.3.2 Real-Adapter Structural Analysis

Experiment 5: Pre-alignment adapter structural analysis. *Aim.* Characterise the structural properties of the five trained QLoRA adapters before any merging or alignment. This is not a test of any method; it is a *diagnostic* experiment whose purpose is twofold: to motivate the scale-aware enhancements of Section 3.2.7 empirically, and to establish the real-adapter baseline against which alignment quality is measured in Experiment 10. *Protocol.* Train five rank-16 QLoRA adapters on SST-2, MNLI, QNLI, CoLA, and RTE using Qwen2.5-1.5B, with per-task hyperparameters tuned to bring each adapter’s standalone performance within published bounds (details in Section 3.4). For each adapter extract the 196 LoRA modules and record per-module Frobenius norms of A_i and B_i , attention-vs-MLP module family breakdowns, per-layer heatmaps, and depth profiles. *Supports.* **C3** (empirical foundation: scale imbalance is real and structured). *Reads as.* If adapter norms differ substantially across tasks, and if the differences are structured by module family and depth rather than uniformly distributed, no single global hyperparameter can rebalance contributions. The scale-aware enhancements of Section 3.2.7 are designed against this diagnosis.

3.3.3 Main Empirical Comparison

Experiment 6: Hyperparameter sweep. *Aim.* Compare methods at *each method’s best* hyperparameters rather than at a shared arbitrary setting. A preliminary comparison at $\lambda = 1$ was misleading because λ is the dominant hyperparameter for both delta-space and factor-space methods, and the two families optimise at different values. *Protocol.* Evaluate every method from Section 3.2 on the five GLUE tasks. Two λ grids are used. For delta-space methods (Task Arithmetic, raw TIES, DARE+TIES): $\lambda \in \{0.1, 0.2, \dots, 1.0\}$, 10 values. For factor-space methods (GPA+TIES and its enhanced variants, LR-KnOTS+TIES): $\lambda \in \{0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5, 0.7, 1.0\}$, 10 values, refined at the low end because preliminary evaluation losses suggested a lower optimum in factor space. Additionally, the TIES trim percentage k is swept over $\{10\%, 20\%, 30\%\}$, the DARE drop probability p over $\{0.1, 0.3, 0.5\}$, and the inverse-norm B -weighting exponent α over $\{0, 0.5, 1.0\}$. *Supports.* **C4** (primary empirical test) and **C5** (the sweep makes CoLA’s collapse observable across all λ). *Reads as.* The main results table in Chapter 4 reports each method at its best configuration; any observed differences are therefore differences between optimally-tuned implementations rather than differences in default settings.

Experiment 7: Three-seed statistical-significance runs. *Aim.* The single-seed gain of the enhanced GPA pipeline over raw TIES in Experiment 6 is small; without variance bars it is unclear whether this reflects a genuine signal or seed-level noise. *Protocol.* Rerun each

of the three primary methods (TIES, LR-KnOTS+TIES, and GPA+TIES) together with the best enhanced variant $\text{dGPA}+\text{saTIES}+\text{wB}(0.5)$ at each method’s best λ from Experiment 6, under three independent random seeds affecting evaluation-time data order and any stochastic tie-breaking within TIES. Report mean $\pm$ standard deviation per task and on the average metric. *Supports.* **C4** as it calibrates the honest margin of the headline result. *Reads as.* If the enhanced-GPA gain lies within one standard deviation of zero, that is reported directly and treated as evidence for C2/C3 rather than as a failure.

Experiment 8: CoLA prediction-distribution diagnostic. *Aim.* The universal CoLA collapse observed in the preliminary comparison has two candidate explanations: a scaling issue (merged weights moving too far from the pretrained solution) or a structural limit (compact merging fundamentally incompatible with the CoLA decision boundary). This experiment separates them. *Protocol.* During the hyperparameter sweep of Experiment 6, record the predicted-class distribution for CoLA at every (λ, k) configuration for every method. Identify whether any configuration produces non-degenerate predictions, and whether the scale-aware enhancements recover CoLA at a lower λ than the baseline. *Supports.* **C5** (primary evidence) and **C3** (scale as binding constraint). *Reads as.* If CoLA collapses at every λ for every method, C5 is established as a structural finding rather than a hyperparameter artefact.

3.3.4 Targeted Ablations

The four ablations reported below each test a specific claim and were selected after the main comparison clarified which questions remained open.

Experiment 9: Enhancement ablation. *Aim.* Isolate which of the three scale-aware enhancements contributes to any gain over the baseline GPA+TIES. *Protocol.* Tabulate the five GPA+TIES variants from Section 3.2.7, each at its best λ from Experiment 6. The individual contributions are read off as differences: $\text{dGPA}+\text{TIES}$ minus GPA+TIES isolates the effect of directional alignment; $\text{dGPA}+\text{saTIES}$ minus $\text{dGPA}+\text{TIES}$ isolates the effect of scale-aware TIES; and $\text{dGPA}+\text{saTIES}+\text{wB}(\alpha)$ minus $\text{dGPA}+\text{saTIES}$ isolates the effect of inverse-norm B weighting. *Supports.* **C3** because the enhancements address scale bias specifically, their relative contributions are a direct diagnostic of whether scale or rotation is the binding constraint. *Reads as.* If only the B -weighting step helps, alignment is not the bottleneck, which is precisely the C3 framing.

Experiment 10: CKA before and after alignment. *Aim.* Measure whether GPA actually increases representational similarity between adapters on real data, not just on synthetic inputs with known rotations. This is the real-adapter counterpart of Experiments 1 and 4. *Protocol.* Compute pairwise Centered Kernel Alignment (Kornblith et al., 2019) between the A_i matrices of all adapter pairs, before alignment (raw A_i) and after alignment (aligned $\tilde{A}_i = Q_i A_i$). Report the two CKA matrices as a heatmap pair. *Supports.* **C1** (GPA works on real data too) and **C2** (even if CKA increases, downstream merge quality may not). *Reads as.* A clear CKA increase accompanied by only modest downstream improvement is the sharpest direct evidence for C2.

Experiment 11: N -ablation (merging 2 to 5 adapters). *Aim.* Test the central multi-way hypothesis introduced in Chapter 1: the benefit of simultaneous alignment should grow with the number of adapters being aligned, because more adapters mean more rotational

disagreement to reconcile. *Protocol.* For each $N \in \{2, 3, 4, 5\}$, evaluate GPA+TIES, the best enhanced variant, TIES, and LR-KnOTS+TIES at each method’s best λ from Experiment 6. For $N = 2$ evaluate all ten pairs; for $N = 3, 4$ evaluate representative subsets; for $N = 5$ evaluate the full set. Report the average primary score as a function of N . *Supports.* **C2** (separability) and **C4** (GPA’s competitive position). *Reads as.* A GPA advantage that grows with N would confirm the multi-way hypothesis; a flat advantage would challenge it directly.

Experiment 12: Task Arithmetic in aligned space. *Aim.* The most direct possible test of C2. If alignment is the binding constraint, applying it to a method as unsophisticated as Task Arithmetic should improve it substantially; if scale is the binding constraint, alignment on its own should change little. *Protocol.* Align with GPA exactly as in GPA+TIES, then merge by summing the reconstructed deltas $\tilde{B}_i \tilde{A}_i$ (Task Arithmetic-style) instead of running TIES. Evaluate at the best λ for Task Arithmetic from Experiment 6. *Supports.* **C2** (direct test). *Reads as.* A substantial improvement would falsify the “scale is the binding constraint” framing; a negligible improvement would strengthen it.

Experiment 13: Per-layer alignment residual and norm correlation. *Aim.* Link the pre-alignment structural diagnosis of Experiment 5 to the post-alignment geometry of GPA on real adapters. If scale imbalance is the binding constraint (C3), the modules in which GPA leaves the most residual disagreement should be those in which the per-adapter B -norm spread is largest, concentrated in the early MLP layers flagged by Experiment 5. *Protocol.* For each of the 196 LoRA modules, record the final alignment residual $\frac{1}{N} \sum_i \|Q_i A_i - C\|_F^2$ and the per-module B -norm spread $\max_i \|B_i\|_F / \min_i \|B_i\|_F$ from the GPA run of Experiment 6 at its best λ . Report the Spearman correlation between the two, stratified by module family, and visualise as a depth $\times$ family residual heatmap. *Supports.* **C1** (on real adapters) and **C3** (directly). *Reads as.* A positive correlation concentrated in early MLP layers is the sharpest real-adapter evidence for C3. A flat correlation weakens C3 and is reported directly, leaving Experiments 5 and 9 as the remaining support.

3.4 Evaluation Protocol and Reproducibility

Base model and adapters. All experiments use Qwen2.5-1.5B as the base model with QLoRA (Dettmers et al., 2023): the frozen base weights are held in 4-bit NF4 quantisation and the trainable LoRA factors are held in higher precision. The adapter rank is $r = 16$ throughout. LoRA modules are placed across the attention and MLP modules of every transformer layer, producing 196 LoRA modules per adapter.

Datasets. The evaluation uses a five-task subset of GLUE: SST-2, MNLI, QNLI, CoLA, and RTE. Training splits are used for adapter training; validation splits are used for all evaluation-time measurements reported in Chapter 4.

Metrics. The primary metric is accuracy for SST-2, MNLI, QNLI, and RTE, and Matthews correlation for CoLA; the latter is chosen because the class imbalance of CoLA renders accuracy uninformative. The average primary score across tasks is reported as the arithmetic mean of the five per-task primary metrics.

Success criterion. The enhanced GPA+TIES pipeline is required to improve the average primary score over the strongest unaligned baseline by at least one percentage point, consistently across at least three of the five tasks. Chapter 4 reports against this threshold honestly, including in cases where the criterion is not met.

Hardware and reproducibility. All training and evaluation runs use a single NVIDIA RTX 6000 Ada (48 GB) with CUDA 12.4. Merging is performed on CPU in NumPy; adapter evaluation is performed on the GPU. Seeds are fixed per run; the three-seed spread of Experiment 7 is archived alongside the experimental results.

Chapter 4

Results and Analysis

Chapter 5

Discussion and Conclusions

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